

Project Work Assignment

Data Analysis, Artificial Intelligence, Machine Learning, and Power BI (30%)

Objective

The objective of this project is to apply Data Analysis, Data Visualization, Data Preprocessing, Feature Engineering, Machine Learning, Artificial Intelligence, and Power BI concepts to solve a real-world problem using a publicly available dataset.

Participants will demonstrate their technical skills, analytical thinking, problem-solving ability, and presentation skills through a project report, dashboard development.

Project Scenario

Select a unique dataset related to your:

- Current Job
- Career Interest
- Business Domain
- Engineering Field
- Accounting and Finance
- Healthcare
- Education
- Marketing
- Banking
- E-commerce
- Agriculture
- Government Sector

Dataset Sources

Choose datasets from:

- Kaggle
- UCI Machine Learning Repository
- Government Open Data Portals

- WHO
- World Bank
- UN Data Repository
- Use your own data
- Other Authentic Open Data Sources

Note: No two students can use the same dataset.

Task 1: Dataset Understanding and Exploration (10 Marks)

Load the dataset using:

- Jupyter Notebook
- Google Colab
- VS Code

Perform:

Basic Information (Any FIVE of the following techniques.)

- Number of Rows
 - Number of Columns
 - Shape
 - Head()
 - Tail()
 - Info()
 - Describe()
 - Data Types
 - Numerical Features
 - Categorical Features
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Task 2: Descriptive Statistical Analysis (10 Marks)

Calculate and interpret: (Any FIVE of the following techniques.)

- Mean
- Median
- Mode

- Maximum
 - Minimum
 - Range
 - Variance
 - Standard Deviation
 - Quartiles
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Task 3: Exploratory Data Analysis (EDA) (10 Marks)

Perform data exploration and visualization.

Data Cleaning

- Missing Values Analysis
- Duplicate Values
- Outlier Detection
- Data Type Conversion
- Inconsistent Data Handling

Demonstrate Pandas Functions (Any FIVE of the following techniques.)

- `append()`
- `concat()`
- `drop()`
- `insert()`
- `rename()`
- `loc[]`
- `iloc[]`
- `groupby()`
- `sort_values()`

Visualizations (10 Marks)

Create at least: (Any Three of the following techniques.)

- Histogram
- Bar Chart
- Scatter Plot
- Box Plot
- Line Chart

- Correlation Heatmap
 - Pair Plot
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Task 4: Data Preprocessing (10 Marks)

Perform preprocessing techniques: (Any two of the following techniques.)

Missing Value Treatment

- Mean Imputation
- Median Imputation
- Mode Imputation

Outlier Handling

- IQR Method
- Z-Score Method

Encoding

- Label Encoding
- One Hot Encoding

Scaling

- Standardization
- Normalization

Data Transformation

- Log Transformation
- Feature Scaling

Data Splitting

- Training Dataset
 - Testing Dataset
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Task 5: Feature Engineering (10 Marks)

Perform feature engineering activities. (Any one of the following techniques.)

Include:

- Feature Creation
- Feature Selection
- Feature Transformation
- Feature Extraction
- Feature Importance Analysis

Apply:

- Correlation Analysis
 - Chi-Square Test
 - PCA
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Task 6: Machine Learning Model Development (10 Marks)

Choose an appropriate machine learning model. (Any one of the following techniques.)

Classification Models

- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- SVM
- Naive Bayes

Regression Models

- Linear Regression
- Polynomial Regression
- Random Forest Regressor

Clustering Models (Optional)

- K-Means

- Hierarchical Clustering

Required Activities

- Model Training
 - Model Testing
 - Model Comparison
 - Best Model Selection
 - Justification of Selected Model
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Task 7: Model Validation and Performance Evaluation (5 Marks)

Evaluate model performance. (Any one of the following techniques.)

Classification Metrics

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score

Regression Metrics

- MAE
 - MSE
 - RMSE
 - R^2 Score
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Task 8: Power BI Dashboard Development (15 Marks)

Create an interactive Power BI dashboard using your dataset.

Dashboard Requirements

Include at least:

- KPI Cards

- Bar Chart
- Line Chart
- Pie/Donut Chart
- Slicers/Filters
- Trend Analysis
- Summary Dashboard

Deliverable

Provide screenshots and explain the dashboard insights.

Task 9: AI-Based Insights and Business Recommendations using Prompt Engineering (10 Marks)

Objective

Choose one business problem or topic from your dataset and apply Prompt Engineering techniques to generate AI-based insights, recommendations, and decision-making support.

Requirements

Apply the following Prompt Engineering Framework:

C4 Framework

- Clarity
 - Context
 - Constraints
 - Format
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Prompt Engineering Activities

Select one business problem from your dataset and create prompts using any FIVE of the following techniques.

- 1. One-Shot Prompting**
- 2. Few-Shot (N-Shot) Prompting**
- 3. Chain-of-Thought Prompting**

4. Persona Pattern

Contextual statements:

- Act as Persona *X*.
- Perform task *Y*.

5. Audience Persona Pattern

Contextual statements:

- Explain *X* to me.
- Assume that I am Persona *Y*.

6. Recipe Pattern

Contextual statements:

- I would like to achieve *X*.
- I know that I need to perform steps *A*, *B*, *C*.
- Provide a complete sequence of steps for me.
- Fill in any missing steps.
- (Optional) Identify any unnecessary steps.

7. Template Pattern

Contextual statements:

- I am going to provide a template for your output.
 - *X* is my placeholder for the content.
 - Try to fit the output into one or more of the placeholders that I list.
 - Please preserve the formatting and overall template that I provide.
 - This is the template: *PATTERN* with *PLACEHOLDERS*.
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Presentation (5–7 Minutes): Prepare a professional presentation: (20%)

Slide 1

Project Title and Participant Introduction

Slide 2

Problem Statement and Dataset Overview

Slide 3

Descriptive Analysis Findings

Slide 4

EDA and Data Visualization

Slide 5

Data Preprocessing and Feature Engineering

Slide 6

Machine Learning Model and Results

Slide 7

Confusion Matrix and Performance Evaluation

Slide 8

Power BI Dashboard Demonstration

Slide 9

Business Insights and Recommendations

Slide 10

Prompt Engineering

Slide 11

Conclusion and Future Scope

